

◦ Aim:

- To determine the pK_a and K_a values of a weak acid by pH titration method.

◦ Principle:

- Weak acid is a weak electrolyte and undergoes only partial dissociation in the solution. For example, acetic acid when dissolved in water,



The dissociation constant, $K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$, $pK_a = -\log(K_a)$.

- Hence, the higher the value of pK_a , lower is the K_a value of the acid and therefore weaker is the acid. pK_a value of a weak acid can be determined by pH titration method. In this method, a known value of the weak acid is titrated with a strong base, and pH of the solution is measured after each small addition of the base. During the titration, the pH of the solution increases gradually till very near to the equivalence point. After the equivalence point, the increase in pH is again very gradual. Therefore the equivalence point is indicated by a sharp increase in pH. The equivalence point is determined by plotting volume of the base against $\Delta pH/\Delta V$, as indicated by the volume corresponding to the maximum of the curve.
- Once the titration is started, the solution contains a weak acid and also its salt with a ~~strong~~ base. Thus the mixture is a buffer mixture. The pH of such a mixture is given by Henderson equation:
$$pH = pK_a + \log \left(\frac{[Salt]}{[Acid]} \right)$$

- At half equivalence point, 50% of the acid is converted to salt, and hence $[acid] = [salt] \Rightarrow pH = pK_a$.

Therefore at half equivalence point, pH gives the pK_a value of weak acid.

Procedure:

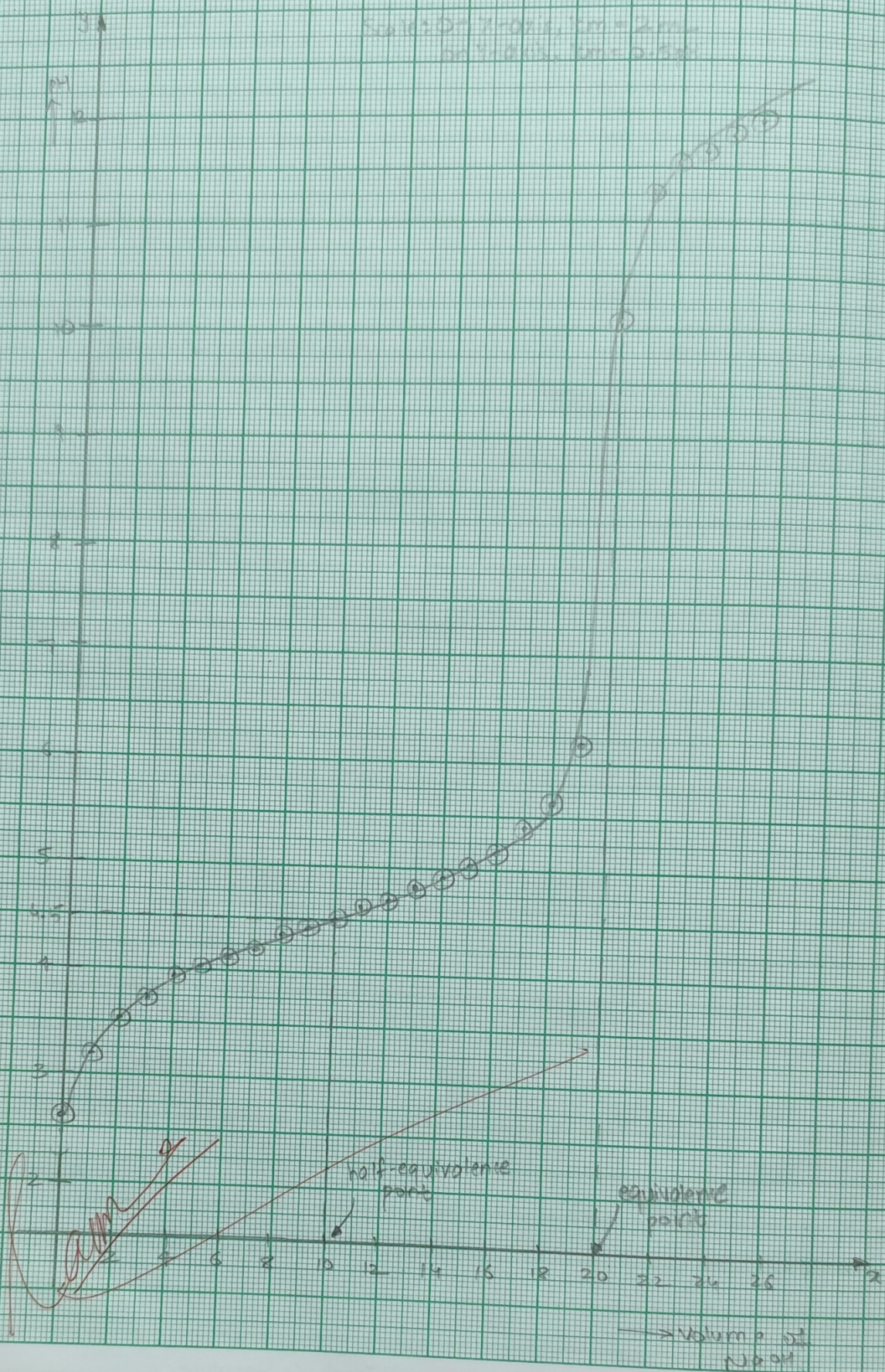
- Pipette out 25 mL of given weak acid into a beaker.
- Dip the glass electrode - calomel electrode assembly into the acid and connect the cell to a pH meter. Measure the pH of the solution.
- Add 1 mL of NaOH at a time into the beaker, stir the solution and measure the pH of the solution.
- Continue the process till a large increase in pH is observed at the end point for the addition of 1 mL of NaOH solution.
- Take four more readings after that.
- Use the readings to plot a graph of volume of NaOH (x-axis) against pH (y-axis).
- Now, clean the beaker and pipette out 25 mL of the weak acid.
- Add 0.5 mL less than the lower range of the end point directly and then 0.1 mL of NaOH at a time.
- After each addition, stir the solution and measure the pH of the solⁿ.
- Continue till a sharp increase in pH is observed for the addition of 0.1 mL of NaOH at the endpoint.
- Continue the titration further to note down at least 4 more readings.
- Plot volume of NaOH (x-axis) against $\Delta pH / \Delta V$ (y-axis) and determine the equivalence point.
- From the graph of volume of NaOH against pH, determine the value of pK_a as pH at half equivalence point.

o observations:

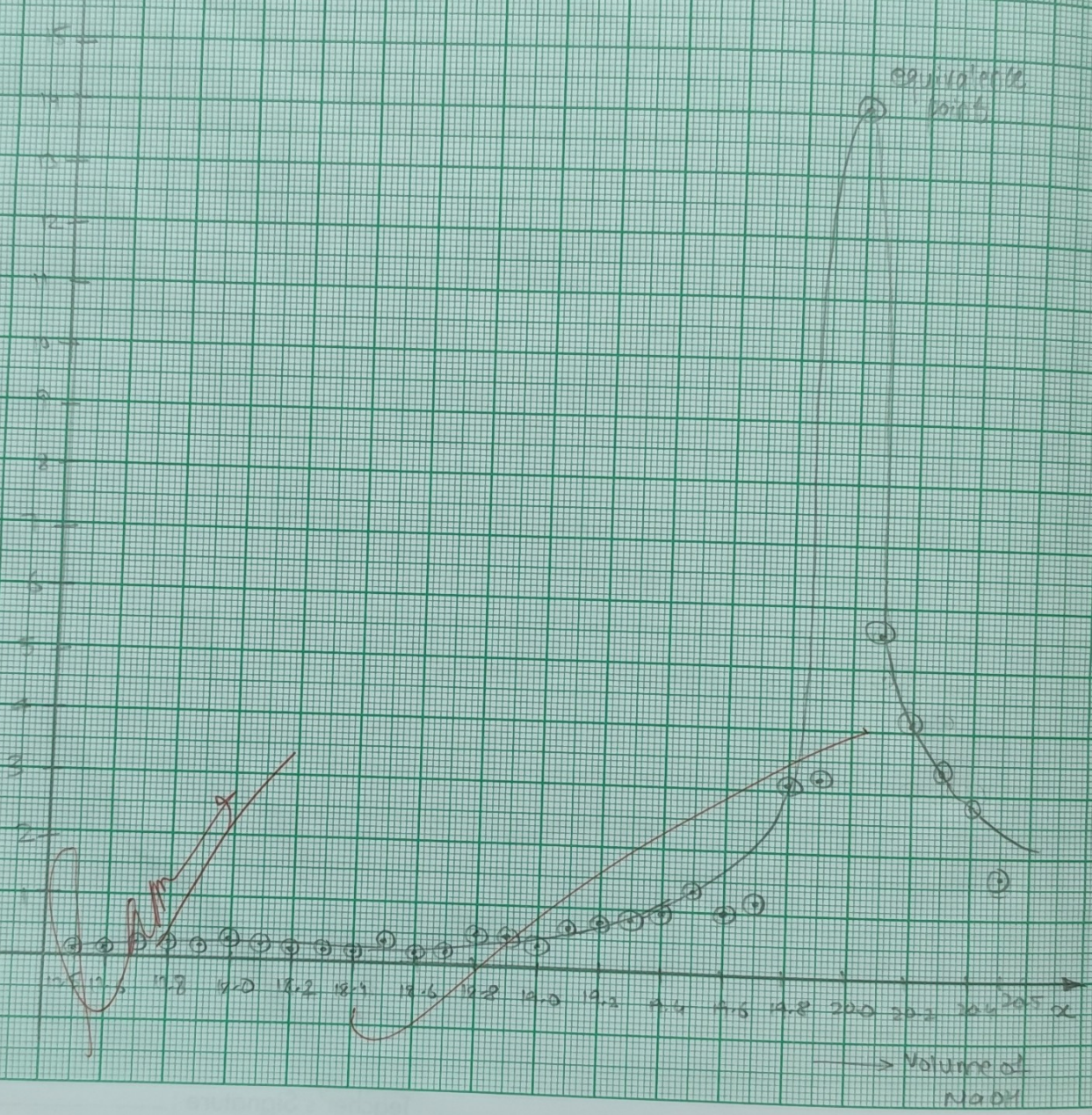
Vol. of NaOH(ml)	pH	Vol. of NaOH(ml)	pH	Vol. of NaOH(ml)	pH
0	2.62	9	4.40	18	5.66
1	3.23	10	4.50	19	6.20
2	3.54	11	4.59	20	10.10
3	3.74	12	4.69	21	11.26
4	3.90	13	4.79	22	11.53
5	4.01	14	4.91	23	11.68
6	4.12	15	5.03	24	11.77
7	4.22	16	5.18	25	11.94
8	4.32	17	5.38		

Vol. of NaOH(ml)	pH	ΔpH	ΔV	$\frac{\Delta pH}{\Delta V}$	Vol. of NaOH(ml)	pH	ΔpH	ΔV	$\frac{\Delta pH}{\Delta V}$
17.5	5.49	-	-	-	19.1	6.00	0.07	0.1	0.7
17.6	5.50	0.01	0.1	0.1	19.2	6.08	0.08	0.1	0.8
17.7	5.51	0.01	0.1	0.1	19.3	6.17	0.09	0.1	0.9
17.8	5.54	0.03	0.1	0.3	19.4	6.27	0.10	0.1	1.0
17.9	5.57	0.03	0.1	0.3	19.5	6.41	0.14	0.1	1.4
18.0	5.60	0.03	0.1	0.3	19.6	6.51	0.10	0.1	1.0
18.1	5.64	0.04	0.1	0.4	19.7	6.62	0.11	0.1	1.1
18.2	5.67	0.03	0.1	0.3	19.8	6.93	0.31	0.1	3.1
18.3	5.69	0.02	0.1	0.2	19.9	7.25	0.32	0.1	3.2
18.4	5.71	0.02	0.1	0.2	20.0	8.66	1.41	0.1	14.1
18.5	5.75	0.04	0.1	0.4	20.1	9.15	0.49	0.1	4.9
18.6	5.77	0.02	0.1	0.2	20.2	9.45	0.30	0.1	3.0
18.7	5.79	0.02	0.1	0.2	20.3	10.00	0.55	0.1	5.5
18.8	5.84	0.05	0.1	0.5	20.4	10.42	0.42	0.1	4.2
18.9	5.89	0.05	0.1	0.5	20.5	10.58	0.16	0.1	1.6
19.0	5.93	0.04	0.1	0.4	-	-	-	-	-

$\text{Sub: } 0.1 \text{ M Fe}^{2+}, V_m = 20 \text{ mL}$
 $\text{or } 0.05 \text{ M, } V_m = 20 \text{ mL}$



Sol: 1. 0.1 g/L, 100 = 0.2 mL
 or 1.0 g/L, 100 = 1.0 mL



= Result:

- From the graph,

- i) Equivalence point of the titration = 20 mL
- ii) pH at half equivalence point = 4.5
- iii) pKa value of given weak acid = 4.5
- iv) Ka of the given weak acid = 3.162×10^{-5}